

Fraud and Fraud Detection Systems

Within the financial services sector, and more specifically, the banking sector, fraud has always been a major challenge, occasioning loss of customer data, unauthorised access to systems and bank accounts, and money. More specifically, in the recent past, the financial services sector has undergone rapid digital transformation, characterised by increased adoption and use of online banking, mobile banking, digital wallets, and electronic payment systems, trends that have significantly increasing the volume and complexity of financial transactions (Gallegos et al., 2025; Wang, 2025). Accordingly, in line with these changes in the financial services sector, there has been a corresponding increase in the reported cases of cyber-enabled fraud, which results in the increased exposure of financial institutions and customers to growing financial and operational risks (Khiaonarong & Zheng, 2026; BioCatch, 2026). In the increasingly digital environment of the financial services sector, fraud includes credit card fraud, account takeover, identity theft, phishing, social engineering, online payment fraud, money laundering, synthetic identity fraud, insider fraud, and authorised push payment (APP) scams (Faisal et al., 2024; Khatib, 2024). Notably, and as Khiaonarong and Zheng (2026) and Hernandez Aros et al. (2024) observed, in the modern environment, fraud has evolved from isolated opportunistic attacks to organised, technology-driven criminal activities that exploit artificial intelligence, malware, automation, botnets, and sophisticated social engineering techniques.

With growing cases and incidents of fraud, institutions and customer are facing extensive financial losses, identity theft, reduced confidence in digital banking, service disruptions, and declining trust in financial institutions, ultimately influencing the adoption of digital financial services (BioCatch, 2026; Wang, 2025). In the context of , studies and reports have highlighted the fact that in the face of growing cases and complexity of fraud, financial institutions continue to experience substantial direct losses and indirect costs associated with investigations, regulatory penalties, operational disruptions, cybersecurity investments, and reputational

damage (Vanini et al., 2023; Bello et al., 2023). As such, to stem the growing negative effects of fraud in the financial services sector, it has become increasingly necessary for firms to design and implement fraud detection systems in response to the changes in fraud patterns. In fact, as Prabowo (2016) and Vanini et al. (2023) observed, fraud detection has evolved from the traditional rule-based approaches to intelligent systems capable of learning complex fraud behaviours from large transaction datasets. Studies show that in the face of evolving fraud and fraud patterns, contemporary fraud detection systems leverage and use various statistical analysis, machine learning, deep learning, anomaly detection, behavioural analytics, graph analytics, and natural language processing (Compagnino et al., 2025; Boulieris et al., 2024). In agreement to the observations by Compagnino et al. (2025) and Boulieris et al. (2024), AbouGrad and Sankuru (2025) observed that current systems are increasingly incorporating real-time monitoring, adaptive learning, explainable AI, and privacy-preserving machine learning to improve effectiveness. However, despite the technological advancements in fraud detection systems, key challenges, including scalability, false positives, concept drift, and computational complexity inhibit the sustainable implementation of fraud detection system (Faisal et al., 2024). These trends and patterns, as demonstrated in the literature, show a gravitation toward integrated, multi-layered fraud detection architectures rather than reliance on single detection techniques.

Traditional and Legacy Fraud Detection Systems

Fraud, as demonstrated, has always been a major challenge in the financial services sector, including the in the pre-technological revolution eras. Commensurately, financial institutions have always developed and implemented measures to combat fraud through fraud detection, prevention, and management. The earliest fraud detection systems, the legacy and traditional fraud prevention mechanisms, were rule-based monitoring systems, manual reviews, and threshold-based detection, which according to Prabowo (2016) and Lim et al. (2021), have

become progressively less effective because they cannot adapt rapidly to emerging fraud patterns and increasingly sophisticated criminal strategies. More specifically, according to Lim et al. (2021), the legacy or traditional fraud detection systems rely primarily on manually developed business rules, expert systems, threshold analysis, blacklists, and statistical techniques. However, according to Achary and Shelke (2023), the legacy or traditional systems are transparent, relatively inexpensive to implement, and easily understood by fraud investigators. Despite the strengths though, as Prabowo (2016) pointed out, the legacy and traditional systems require manual updating and cannot effectively detect previously unseen fraud patterns. Further, as Vanini et al. (2023) pointed out, another notable weakness of legacy systems is that they generate high false-positive rates frequently and struggle with large-scale online transaction environments. To sum up the nature and functionality of legacy and traditional fraud detection systems, Hernandez Aros et al. (2024) observed that they are generally complementary screening mechanisms, instead of comprehensive fraud detective solutions.

Machine Learning-Based Fraud Detection Systems (Sanni/Omatayo)

With technological innovations and changes, organisations in the financial services sectors have increasingly adopted and used the emergent technologies in fraud detection, such as machine learning (ML). Machine learning-based systems have been one of the key technology-based fraud detection systems, which leverage a combination of technologies to automatically identify hidden fraud patterns from historical transaction data and continuously improve predictive performance (Compagnino et al., 2025). More specifically, according to Lim et al. (2021) and Islam et al. (2024), in ML, supervised learning has remained as one of the dominant approaches, but organisations are increasingly adopting and using unsupervised, semi-supervised, deep learning, and ensemble methods to detect current and emergent fraud schemes (Lim et al., 2021; Islam et al., 2024). Further, as Khalid et al. (2024) and Chen (2025) noted,

ensemble algorithms such as Random Forest, XGBoost, Gradient Boosting, and LightGBM consistently outperform many traditional ML algorithms across multiple evaluation metrics. Further, within the context of ML, different studies, as Albalawi and Dardouri (2025) studies have showed that deep learning models have high predictive performance, but the computational resources requirements and limited interoperability inhibits their widespread adoption and use in the sector.

From another perspective, other scholars, such as Luong and Xie (2026) and in another article, AbouGrad and Sankuru (2025) highlighted the gravitation toward explainable AI and privacy-preserving machine learning, which emerged as an important research direction, with an objective of improving transparency and regulatory compliance. Similarly, Compagnino et al. (2025) and Dichev et al. (2025) observed that recent advancements in AI and ML have transformed fraud detection by enabling financial institutions to identify complex behavioural patterns, detect anomalies, and therefore, automate fraud risk assessment with greater accuracy than conventional systems. However, despite these advancements and trends, Al-dahasi et al. (2025) and in another study, Zafar and Wu (2026) pointed out that fraud detection remains constrained by significant challenges, including class imbalance, concept drift, adversarial attacks, high false-positive rates, limited explainability, data privacy concerns, and increasing regulatory requirements. Summing up current trends and practices, AbouGrad and Sankuru (2025) and others, such as Luong and Xie (2026) observed that organisations in the sector should focus on developing adaptive, explainable, privacy-preserving, and real-time machine learning frameworks capable of responding to evolving fraud patterns while maintaining regulatory compliance and customer trust (AbouGrad & Sankuru, 2025; Luong & Xie, 2026). Therefore, critically, although ML has substantially improved fraud detection performance, current evidence indicates that no universally optimal algorithm exists, reinforcing the need for context-specific model evaluation and hybrid detection frameworks.